

Lecture 04

Data and Analysis

Chapter lecture — 8 topics with explanations, worked examples, Urdu notes, practice questions and a full answer key.

BIEK Computer Science XII · Bilingual support: English + اردو

بی آئی ای کے کمپیوٹر سائنس — بارہویں جماعت

08 book lectures in this pack

★ Golden topics, Urdu notes, review and answer key

XII

How to use this BIEK Computer Science lecture

Each PDF is a classroom lecture from the BIEK / Sindh Computer Science textbooks (New Sindh Curriculum, STBB Grade XI–XII). Read *Learn it*, study the SVG diagram, try the worked example on paper, then answer *Check yourself* before looking at the answers. **Extra board practice** (MCQs, short and long questions) is added after the book content — it is not in the textbook. Career-only extras that were not in the book lecture list have been removed. Math is rendered with MathJax. ★ Golden topics are high-yield in board exams. The green box repeats the idea in Urdu.

ہر لیکچر بی آئی کے / سندھ نصاب کی کتاب کے درس کے مطابق ہے۔ سبق پڑھیں، مثال حل کریں، پھر چیک یورسیلف کے سوالات کے جواب اپنے نوٹ بک میں لکھیں۔ کتاب کے بعد اضافی بورڈ پریکٹس ہے۔ ★ گولڈن ٹاپکس امتحان میں زیادہ پوچھے جاتے ہیں۔

Lectures in this pack

Chapter 4: Data and Analysis. Study one lecture at a time, then use the chapter review and answer key at the end of this PDF.

L01. 4.1 Concept of Data Analysis

L02. 4.1.1 – 4.1.2 Data Sources and Database Connection

L03. 4.2 Creating a SQLite Database in Python

L04. 4.3 Basic Operations on Data using Pandas

L05. 4.3.3 Handling Missing Values (NaN)

L06. 4.4 Data Organization and Representation

L07. 4.5 Data Visualization & Types of Graphs

L08. 4.6 Descriptive Statistics

Data and Analysis

This chapter shows you how to pull data from a SQLite database with Python, clean it in Pandas, draw the right chart, and summarise a dataset with mean, median, mode and standard deviation. These are the same steps a data analyst uses every day.

اس باب میں آپ سیکھیں گے کہ پائتھن سے ایس کیو لائٹ ڈیٹا بیس کیسے جوڑا جاتا ہے، پینڈاز میں ڈیٹا کیسے صاف ہوتا ہے، درست چارٹ کیسے بنتا ہے، اور اوسط، میڈین، موڈ اور معیاری انحراف سے خلاصہ کیسے نکلتا ہے۔

What you will learn

- Define data analysis and say why Python is used for it.
- Create a SQLite database, table, insert, select, commit and close.
- Create calculated columns and handle NaN with isnull, fillna and dropna.
- Choose and code bar, line, pie, histogram, scatter and box plots.
- Name the five parts of a Python database connection in order.
- Build a Pandas DataFrame from a dictionary, CSV or Excel file.
- Organise data in tables and frequency distributions.
- Compute mean, median, mode, range, variance, standard deviation, min and max by hand, and know that pandas `.std()` uses `n-1`.

4.1 Concept of Data Analysis

LEARN IT

Data analysis is examining, organising and interpreting data so you can extract useful information. Schools study results, shops study sales, hospitals study patient records — all to make a decision, spot a trend or solve a problem.

Python is popular for this work because it is readable, talks to files and databases, does statistics, and draws graphs. Five lines of Pandas often replace an hour of spreadsheet clicking.

WORKED EXAMPLE

A Hyderabad college has 400 CS marks in a spreadsheet. Analysis answers: What is the average? How many failed? Is the second shift weaker than the first? The numbers are data; those answers are information.

اردو میں سمجھیں

ڈیٹا کا تجزیہ یعنی اعداد کو ترتیب دے کر ان سے فیصلے کے قابل معلومات نکالنا۔ پائٹھن اس لیے منتخب کیا جاتا ہے کہ وہ فائلیں، ڈیٹا بیس، حساب اور گراف سب سنبھال لیتا ہے۔

CHECK YOURSELF

1. Give one school use of data analysis.
2. Name two Python superpowers for analysis.

Answers: 1. Study results to improve teaching (or similar). 2. Any two: read CSV/Excel/databases, organise tables, statistics, charts.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Data analysis is:

- a. Examining data to extract useful information
- b. Only collecting rumours
- c. Only drawing logos
- d. Only cabling

MCQ 2. Python is used because:

- a. Files, databases, stats, plots in one language
- b. It is analog
- c. It cannot loop
- d. It cannot import

MCQ 3. Raw marks vs "class average 62":

- a. Data vs information
- b. Information vs data
- c. Both keys
- d. Both NAND

MCQ 4. A decision is the:

- a. Point of analysis
- b. First NAND
- c. Last OSI layer always
- d. Sprint name

MCQ 5. Without a question, analysis is:

- a. Aimless charting
- b. Always AI
- c. Always IoT
- d. Always HCI

MCQ 6. XII tools for analysis include:

- a. sqlite3, pandas, matplotlib, statistics
- b. Only LogiSim
- c. Only Figma
- d. Only Access macros

Extra short questions

1. Define data analysis with a college example.
2. Four Python superpowers listed in the notes.
3. What is not analysis?
4. Link to XI digital literacy.

Extra long question

1. A Hyderabad college has 400 CS marks. Write four analysis questions and which Python tool answers each.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Process marks to decide who needs extra class.
2. Read files, talk to DB, statistics, charts (and readable syntax).
3. Dumping 400 rows on a slide with no summary.
4. Same collect-present loop, now with code.

Long

1. Average: pandas mean. Fail count: filter. Shift compare: groupby. Trend: matplotlib line if by year. Mention cleaning NaNs first.

4.1.1 – 4.1.2 Data Sources and Database Connection ★ Golden

LEARN IT

Data sources are where information comes from: school records, websites, sensors, databases. **Connectivity** is the conversation between your Python program and that source so you can read and update data.

Five components of a database connection, in the order they actually work:

#	Component	Role
1	Database	Organised store (SQLite = one .db file; MySQL = a server for big apps)
2	Driver	Translator between Python and that DBMS (sqlite3 is built in)
3	Connection object	Opens, manages and closes the pipeline
4	Cursor object	Runs SQL and walks through result rows
5	SQL queries	CREATE, INSERT, SELECT, UPDATE, DELETE



EXAM TIP

Write the five names in order and one line each. A common trap is swapping cursor and connection: the connection is the pipe; the cursor is the messenger that travels through it.

اردو میں سمجھیں

خمسہ اجزاء: ڈیٹا بیس، ڈرائیور، کنکشن، کرسر، ایس کیو ایل۔ کنکشن پائپ ہے، کرسر اس پائپ میں چل کر حکم چلاتا ہے۔

CHECK YOURSELF

1. Which object executes `SELECT *`?
2. Is SQLite a driver, a database, or both?

Answers: 1. The cursor. 2. SQLite is the database; the `sqlite3` module is the driver.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. The five connection parts in order start with:

- a. Database then driver
- b. Cursor then database
- c. SQL then mouse
- d. Chart then file

MCQ 2. The cursor:

- a. Executes SQL and fetches rows
- b. Is the .db file
- c. Is matplotlib
- d. Is Figma

MCQ 3. The connection object is the:

- a. Pipeline
- b. SQL text
- c. Driver brand only
- d. Pie

MCQ 4. sqlite3 is the:

- a. Built-in driver (+ DB engine in-process)
- b. A web browser
- c. An OSI layer
- d. A sort

MCQ 5. SQL examples:

- a. SELECT, INSERT, CREATE
- b. def, for, if
- c. AND, OR, NOT gates
- d. HTTP GET only

MCQ 6. Swapping cursor and connection in an answer is a:

- a. Classic 0-mark mix-up
- b. Required trick
- c. Big O class
- d. UX heuristic

Extra short questions

1. Name the five components in order.
2. Why a driver?
3. SQLite vs MySQL in one line.
4. Draw the flow.

Extra long question

1. Label a 9-line sqlite snippet with the five components. Explain commit vs close. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Database, driver, connection, cursor, SQL.
2. Translates Python calls to the DBMS protocol/API.
3. SQLite: file, no server. MySQL: server for many users.
4. Python → driver → connection → cursor → SQL → DB.

Long

1. import=driver; connect=connection; cursor=cursor; execute=SQL;
school.db=database. commit saves; close unlocks. Missing commit → empty next run.

4.2 Creating a SQLite Database in Python

LEARN IT

SQLite is built into Python. The whole database is one `.db` file next to your script — no password, no port, no server. Nine steps you must be able to write from memory:

```
import sqlite3
conn = sqlite3.connect("school.db")    # 1-2  driver + connection
cur = conn.cursor()                   # 3    cursor
cur.execute("CREATE TABLE IF NOT EXISTS students (id INTEGER, name TEXT, marks
INTEGER)")
cur.execute("INSERT INTO students VALUES (1, 'Ali', 85)")
conn.commit()                         # 6    SAVE or the row vanishes
cur.execute("SELECT * FROM students")
rows = cur.fetchall()
for r in rows:
    print(r)
conn.close()                          # 9    unlock the file
```

OUTPUT

```
(1, 'Ali', 85)
```

Step	Why it matters
<code>connect</code>	Creates the file if it does not exist
<code>cursor</code>	Vehicle that carries SQL
<code>execute</code>	Runs one SQL command
<code>commit</code>	Writes INSERT/UPDATE/DELETE to disk
<code>fetchall</code>	Brings SELECT results into a Python list of tuples
<code>close</code>	Frees memory and unlocks the file

COMMON MISTAKE

Forgetting `commit()` is the classic bug: the program prints nothing next time because the insert was never saved. Forgetting `close()` can lock `school.db` so the next run fails.

REMEMBER IT

Strings inside SQL use single quotes: `'Ali'`. Python strings around the SQL use double quotes (or a triple-quoted string).

ایس کیو لائٹ پائٹھن کے ساتھ آتی ہے، پورا ڈیٹا بیس ایک فائل ہے۔ ترتیب: امپورٹ، کنکٹ، کرسر،
COMMIT .EXECUTE، COMMIT، SELECT، FETCALL، CLOSE نہ ہو تو اندراج ضائع ہو جاتا
ہے۔

CHECK YOURSELF

1. What does * mean in SELECT *?
2. Which call must follow INSERT?

Answers: 1. All columns. 2. conn.commit() .

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. `sqlite3.connect("school.db")` will:

- a. Open or create the file
- b. Always error if missing
- c. Start MySQL server
- d. Draw a pie

MCQ 2. After INSERT you must:

- a. `commit()`
- b. only close
- c. drop the table
- d. `pip install sqlite`

MCQ 3. `fetchall()` returns:

- a. A list of tuples (rows)
- b. A DataFrame always
- c. A dict of dicts always
- d. A string of SQL

MCQ 4. `CREATE TABLE IF NOT EXISTS` avoids:

- a. Error on re-run
- b. The need for types
- c. The need for keys
- d. `commit`

MCQ 5. `close()` is needed to:

- a. Unlock the file
- b. Create pandas
- c. Sort rows
- d. Encrypt OSI

MCQ 6. `SELECT *` means:

- a. All columns
- b. All tables in the folder
- c. A pie of all apps
- d. Drop everything

Extra short questions

1. Nine-step sequence from the notes.
2. What does `*` mean?
3. Two classic bugs.
4. How do you see the data next run?

Extra long question

1. Write a complete program: table `students(id, name, marks)`, insert Ali 85 and Sara 90, print rows. Number the five connection components in comments.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. import, connect, cursor, create, insert, commit, select, fetch, close.
2. Every column.
3. No commit; forgotten close; SQL string quotes mixed up.
4. It is in school.db — connect again and SELECT.

Long

1. Full working code and output two tuples. Comments:
driver/connection/cursor/SQL/database.

4.3 Basic Operations on Data using Pandas ★ Golden

LEARN IT

Pandas is a third-party library for tables. Install once: `pip install pandas` in the terminal (not inside the `.py` file). Convention: `import pandas as pd`.

A **DataFrame** is a 2-D table of rows and columns, like Excel or an SQL table. You can filter, sort and calculate whole columns without writing a `for` loop.

```
import pandas as pd
data = {
    "Name": ["Ali", "Sara", "Ahmed"],
    "Math": [85, 90, 78],
    "Science": [80, 88, 70],
}
df = pd.DataFrame(data)
print(df)
```

OUTPUT

	Name	Math	Science
0	Ali	85	80
1	Sara	90	88
2	Ahmed	78	70

`read_csv('students.csv')` and `read_excel('students.xlsx', sheet_name='Sheet1')` load files. `df.head()` prints the first five rows — use it on large files so you do not flood the screen.

```
df["Total"] = df["Math"] + df["Science"]
df["Average"] = df["Total"] / 2
print(df)
```

OUTPUT

	Name	Math	Science	Total	Average
0	Ali	85	80	165	82.5
1	Sara	90	88	178	89.0
2	Ahmed	78	70	148	74.0

WORKED EXAMPLE

Check: Ali $85+80=165$, $165/2=82.5$. Sara $90+88=178$, 89.0 . Ahmed $78+70=148$, 74.0 .

اردو میں سمجھیں

پینڈا ڈیٹا فریم دیتا ہے۔ ڈکشنری، سی ایس وی یا ایکسل سے لوڈ کریں۔ نئے کالم بغیر لوپ کے پوری قطار پر حساب کر کے بنتے ہیں۔ `head()` صرف پہلی پانچ قطاریں دکھاتا ہے۔

CHECK YOURSELF

1. What is `pd`?
2. Why not `print(df)` on a million-row file?

Answers: 1. The usual alias for pandas. 2. It would print everything; use `df.head()`.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Common DS operations include:

- a. Insert, delete, search, traverse
- b. Only compile
- c. Only ping
- d. Only print the logo

MCQ 2. Traverse means:

- a. Visit elements in some order
- b. Delete all
- c. Encrypt all
- d. Sort the OS

MCQ 3. Search cost depends on:

- a. The structure (array vs sorted vs hash vs tree)
- b. The PDF theme
- c. The teacher name
- d. The sprint

MCQ 4. Insert in a full array may:

- a. Fail or require resizing
- b. Always $O(1)$ no matter what
- c. Create a graph edge automatically
- d. Pop a stack

MCQ 5. Delete from a queue happens at the:

- a. Front
- b. Rear only always
- c. Random index
- d. Root of a tree

MCQ 6. You choose a DS by the:

- a. Operations you will do most
- b. First letter of the course
- c. Colour
- d. NAND count

Extra short questions

1. List five operations.
2. Give costs qualitatively: array index vs list search.
3. What does traverse a tree mean?
4. Why "update" might be search+insert.

Extra long question

1. For array, stack, queue, tree: name the natural insert/delete ends and one costly operation. Table.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Create, insert, delete, search, traverse, update, sort (as applicable).
2. Array index $O(1)$; unsorted list search $O(n)$.
3. Visit every node (pre/in/post or level order).
4. You must find the item first.

Long

1. Array: insert middle costly. Stack: top only cheap. Queue: rear in front out. Tree: search/insert depend on balance. 8-row table.

4.3.3 Handling Missing Values (NaN) ★ Golden

LEARN IT

Missing cells are **NaN** ("Not a Number"). Statistics crash or lie if you ignore them. Three tools: `isnull()` (True where empty), `fillna(value)` (replace), `dropna()` (delete the row).

```
import pandas as pd
df = pd.DataFrame({
    "Name": ["Ali", "Sara", "Ahmed", "Ayesha"],
    "Marks": [85.0, None, 78.0, 92.0],
    "Grade": ["A", "A", None, "A"],
})
print(df.isnull().sum())
```

OUTPUT

```
Name      0
Marks      1
Grade      1
dtype: int64
```

WORKED EXAMPLE

Replace Marks with the mean of the known marks, and Grade with a label. Known marks 85, 78, 92 → mean = $255/3 = 85.0$.

```
avg = df["Marks"].mean()          # 85.0
df["Marks"] = df["Marks"].fillna(avg)
df["Grade"] = df["Grade"].fillna("Not Assigned")
print(df)
```

OUTPUT

	Name	Marks	Grade
0	Ali	85.0	A
1	Sara	85.0	A
2	Ahmed	78.0	Not Assigned
3	Ayesha	92.0	A

`df = df.dropna()` would instead delete Sara's row and Ahmed's row (any row that still had a NaN).

EXAM TIP

State: identify with `isnull / sum`, fill numeric gaps with the mean, fill categories with a label, or drop rows if filling would invent a lie. Never analyse first and clean second.

COMMON MISTAKE

Filling a salary gap with the mean of a company that includes the CEO can hide poverty (the same outlier problem as the mean). Know *why* the value is missing before you fill.

اردو میں سمجھیں

خالی خانے NaN ہیں۔ `isnull` تلاش، `fillna` بھرنا (اکثر اوسط سے)، `dropna` قطار ہٹانا۔ پہلے صاف کرو، پھر تجزیہ۔

CHECK YOURSELF

1. What does `df.isnull().sum()` return?
2. Does `dropna()` delete a column or a row by default?

Answers: 1. Count of NaNs in each column. 2. Any **row** that contains at least one NaN.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. NaN means:

- a. Not a Number — missing
- b. A huge number
- c. Zero always
- d. A pie slice

MCQ 2. `isnull().sum()` gives:

- a. Count of missing per column
- b. The mean
- c. The PK
- d. The chart type

MCQ 3. `fillna(mean)` :

- a. Replaces gaps with the average
- b. Deletes the column names
- c. Sorts
- d. Commits SQL

MCQ 4. `dropna()` by default:

- a. Drops rows that contain any NaN
- b. Drops the whole DataFrame always
- c. Fills with 0
- d. Draws a box plot

MCQ 5. You should clean NaNs:

- a. Before analysing
- b. After publishing
- c. Never
- d. Only in Figma

MCQ 6. Filling salary with the mean can:

- a. Hide poverty if a CEO is in the data
- b. Fix ethics
- c. Create keys
- d. Route packets

Extra short questions

1. Three tools: find, fill, drop.
2. Mean of 85, None, 78, 92 for filling.
3. When drop is better than fill.
4. `isnull()` returns what type of frame?

Extra long question

1. Given Marks [85, None, 78, 92] and Grade [A,A,None,A]: show `isnull` sum, fill marks with mean and grade with "Not Assigned", print the frame. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. `isnull/sum; fillna; dropna`.

2. Known $85+78+92=255/3=85.0$.
3. If inventing a value would lie (unknown category, huge outlier).
4. Booleans True where missing.

Long

1. Counts 1 and 1. Sara marks 85.0, Ahmed grade Not Assigned. Code + table. Warn about dropna deleting two rows if used first.

4.4 Data Organization and Representation

LEARN IT

Organised data is easier to read. Three everyday forms:

- **Table** — exact values side by side (Ali 85 Math, 80 Science).
- **Frequency distribution** — how many items fall in each range (bin).
- **Graph** — a picture of a trend or comparison (next topics).

Range of marks	Frequency (students)
50–59	5
60–69	7
70–79	9
80–89	4
90–100	2

Most students scored 70–79. That pattern is hard to see in a raw list of 27 marks, and obvious in this table. Matplotlib (`import matplotlib.pyplot as plt`) turns tables into graphs: `import` → `figure` → `plot` → `labels` → `title` → `show()`.

اردو میں سمجھیں

ٹیبل درست عدد دکھاتی ہے، تعدد کی جدول بتاتی ہے کتنے طلبہ کس رینج میں ہیں، گراف رجحان دکھاتا ہے۔
میٹپلاٹ لب گراف بناتی ہے۔

CHECK YOURSELF

1. How many students scored 80 or above in the table?
2. Name the library that draws graphs in Python.

Answers: 1. $4 + 2 = 6$. 2. Matplotlib (pyplot).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Tabular form is for:

- a. Exact values in rows/columns
- b. Only a pretty poster
- c. Only NAND
- d. Only sprints

MCQ 2. A frequency table shows:

- a. How many in each bin/range
- b. The ER diagram
- c. The OSI stack
- d. The Figma layers

MCQ 3. Most students in the notes' table scored:

- a. 70–79
- b. 90–100
- c. 0–10
- d. Exactly 100

MCQ 4. Graphical representation is for:

- a. Seeing patterns fast
- b. Storing PKs
- c. Compiling C
- d. Routing

MCQ 5. Matplotlib import line:

- a. `import matplotlib.pyplot as plt`
- b. `import pie`
- c. `import osi`
- d. `import nand`

MCQ 6. `plt.show()` :

- a. Renders the figure
- b. Commits SQL
- c. Closes sqlite
- d. Starts Figma

Extra short questions

1. Three organisation methods in the notes.
2. From the frequency table, how many scored 80+?
3. Steps of a matplotlib figure.
4. Why frequencies for large n?

Extra long question

1. Build a frequency table for 12 marks you invent, then say which graph you would draw and why. Include axis titles.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Table, frequency distribution, graph.
2. $4+2=6$.
3. import, figure, plot, labels, title, show.
4. A list of 2000 marks is unreadable; bins are not.

Long

1. Bins of 10. Most-filled bin noted. Histogram (numeric distribution) or column if you treat bands as categories — say which and why bars touch or not.

4.5 Data Visualization & Types of Graphs ★ Golden

LEARN IT

Pick the graph that matches the question. Then label axes and give a title — an unlabelled chart scores zero in an exam.

Chart	Shows	Watch out
Bar / column	Compare discrete categories (subjects, departments)	Column = vertical; bar often = horizontal
Line	Change over time (temperature, visitors)	Needs an ordered x-axis (months, years)
Pie	Parts of one whole as percentages	Bad with many or almost-equal slices
Histogram	Distribution of one numeric variable in bins	Bars touch (continuous range), unlike a bar chart
Scatter	Relationship of two numeric variables	Each dot is one pair (x, y)
Box plot	Min, Q1, median, Q3, max, and outliers	Best tool for spotting extreme values

4.5.1 Bar / column

```
import matplotlib.pyplot as plt
subjects = ["English", "Math", "Science", "History"]
marks = [64, 82, 71, 49]
plt.bar(subjects, marks, color="skyblue", edgecolor="black")
plt.title("Marks vs Subjects")
plt.xlabel("Subjects")
plt.ylabel("Marks out of 100")
```

4.5.2 Line

```
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
temps = [12, 15, 20, 25, 30, 35]
plt.plot(months, temps, marker="o")
plt.title("Average Monthly Temperature")
plt.grid(True)
```

4.5.3 Pie

```
subjects = ["Math", "Physics", "CS", "Biology"]
students = [25, 15, 30, 10] # total 80
plt.pie(students, labels=subjects, autopct="%1.1f%%", startangle=90)
```

Check: Math $25/80 = 31.25\% \rightarrow 31.2\%$. Physics 18.8% . CS 37.5% . Biology 12.5% .

4.5.4 Histogram

```
scores = [45, 50, 55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 82, 85, 90, 95]
plt.hist(scores, bins=[40, 50, 60, 70, 80, 90, 100], edgecolor="black")
```

4.5.5 Scatter

```
plt.scatter([10, 20, 30, 40, 50, 60, 70, 80],
            [2, 4, 5, 4, 7, 8, 9, 12], color="green", alpha=0.7)
plt.title("Rainfall vs Crop Yield") # positive correlation
```

4.5.6 Box plot

```
plt.boxplot([[55, 60, 65, 70, 75, 80, 85],
            [40, 65, 70, 72, 75, 80, 95]],
            tick_labels=["Class A", "Class B"])
```

The box is the middle 50% (Q1 to Q3). The line inside is the median. Whiskers reach min and max (or a fence, with outliers drawn as dots). Class B is more spread out than Class A.



Column compares; line shows a trend; pie shows share; the box summarises spread around the median.

EXAM TIP

"Which graph and why?" is a 3-mark gift. Write the name, what it shows, and one reason the others are worse.

اردو میں سمجھیں

بار/کالم اقسام کا موازنہ، لائن وقت کا رجحان، پائی حصے، ہسٹوگرام مسلسل تقسیم (بار آپس میں لگے)، اسکیٹر دو متغیرات کا تعلق، باکس پلاٹ پانچ اعداد اور آؤٹ لائر۔

CHECK YOURSELF

1. Why do histogram bars touch?
2. Rainfall vs yield — which chart?

Answers: 1. The x-axis is a continuous numeric range, not separate categories. 2. Scatter (two numeric variables, looking for correlation).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Bar/column charts compare:

- a. Discrete categories
- b. Parts of one whole only
- c. A single time point of one var as a distribution
- d. A 5-number summary

MCQ 2. Line graphs are for:

- a. Trends over time
- b. Unrelated categories with gaps
- c. Shares of a budget
- d. Outliers in five numbers

MCQ 3. Pie charts need:

- a. Parts of one whole; few slices
- b. 60 slices of 1%
- c. Time on x
- d. Two numeric axes of correlation

MCQ 4. Histogram bars:

- a. Touch (continuous bins)
- b. Must have large gaps
- c. Are always time
- d. Replace the mean

MCQ 5. Scatter plots show:

- a. Two numeric variables (correlation)
- b. One category vs colour
- c. FIFO order
- d. OSI layers

MCQ 6. A box plot shows:

- a. Min, Q1, median, Q3, max (+ outliers)
- b. Only the mean
- c. Only mode
- d. Only n

Extra short questions

1. One-line job of each of the six charts.
2. Why histogram $\neq$ bar.
3. When pie fails.
4. Rainfall vs yield?

Extra long question

1. Six school questions $\rightarrow$ pick the chart and a matplotlib function (bar, plot, pie, hist, scatter, boxplot). Justify in one sentence each. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Bar categories; line time; pie share; hist distribution; scatter relationship; box spread/outliers.
2. Numeric bins, touching bars vs separate categories.
3. Too many / nearly equal slices.
4. Scatter; look for positive correlation.

Long

1. Enrolment by faculty: bar. Hits by month: plot. Club shares: pie. Heights: hist. Hours vs marks: scatter. Two classes' spread: boxplot. Unlabelled charts score 0.

4.6 Descriptive Statistics ★ Golden

LEARN IT

Descriptive statistics squeeze a long list into a few honest numbers. **Central tendency** asks “what is typical?”. **Spread** asks “how mixed is the list?”.

4.6.1 Mean, median, mode

Measure	How	Use when
Mean	Sum ÷ n. Uses every value.	Values sit close together. Example: ordinary test scores.
Median	Middle after sorting. Odd n: centre. Even n: mean of the two centre values.	Outliers would wreck the mean (CEO salary).
Mode	Most frequent value. May be none, one, or several.	Most popular choice (shoe size).

WORKED EXAMPLE

Marks 70, 80, 90. Mean = $240/3 = 80$.

Odd list 35, 10, 15, 70, 95. Sort: 10, 15, 35, 70, 95. Median = **35**.

Bottles sold 20, 25, 20, 30, 20, 25, 20. Mode = **20** (appears four times).

Outlier trap: nine people earn 10, one earns 1000. Mean = $1090/10 = 109$ (looks rich).

Median = 10 (tells the truth).

4.6.2 Spread: range, variance, standard deviation, min, max

- **Min / max** — smallest and largest. Marks 70, 80, 90, 60, 75 → min 60, max 90.
- **Range** = max – min. Sales 120, 150, 180 → range 60.
- **Population variance** $\sigma^2 = \frac{1}{n} \sum (x_i - \mu)^2$.
- **Sample variance** s^2 uses $\div (n - 1)$. `pandas.Series.std()` uses the sample version.
- **Standard deviation** $\sigma = \sqrt{\sigma^2}$. Same units as the data (marks, not marks²). Low SD → tight, consistent. High SD → spread out, unpredictable.

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2, \quad \sigma = \sqrt{\sigma^2}, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

WORKED EXAMPLE

Full working on 70, 80, 90. Mean $\mu = 80$. Deviations -10, 0, +10. Squares 100, 0, 100. Sum = 200.

Population variance $\sigma^2 = 200/3 = \mathbf{66.67}$ (2 d.p.). Population SD $\sigma = \sqrt{200/3} = \mathbf{8.16}$ (2 d.p.).

Sample variance $s^2 = 200/2 = \mathbf{100}$. Sample SD $s = 10$. That is what `pd.Series([70,80,90]).std()` returns.

EXAM TIP

Always state whether you used n or $n-1$. Show the sorted list for the median. For SD, show: mean $\rightarrow$ deviations $\rightarrow$ squares $\rightarrow$ divide $\rightarrow$ square root. A box plot pictures min, Q1, median, Q3, max of the same data.

COMMON MISTAKE

Do not report variance as if it were in "marks". Variance is marks^2 . Quote the standard deviation when you want a sentence in the original units.

اردو میں سمجھیں

اوسط سب کا مجموعہ تقسیم تعداد، میڈین ترتیب کے بعد بیچ، موڈ سب سے زیادہ بار آنے والا۔ رینج بڑا منفی چھوٹا۔ ویرینس اوسط سے فاصلوں کے مربع کی اوسط ہے، معیاری انحراف اس کا جذر ہے اور اصل اکائی میں ہوتا ہے۔ پینڈاز std() نمونے والا ($n-1$) استعمال کرتا ہے۔

CHECK YOURSELF

1. Median of 4, 8, 6, 10, 2?
2. Range of 120, 150, 180?
3. Does pandas .std() divide by n or $n-1$?

Answers: 1. Sorted 2,4,6,8,10 $\rightarrow$ 6. 2. 60. 3. $n-1$ (sample).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Mean of 70,80,90 is:

- a. 80
- b. 70
- c. 90
- d. 240

MCQ 2. Median of 10,15,35,70,95 is:

- a. 35
- b. 10
- c. 70
- d. 45

MCQ 3. Mode of 20,25,20,30,20,25,20 is:

- a. 20
- b. 25
- c. 30
- d. 7

MCQ 4. Range of 120,150,180 is:

- a. 60
- b. 30
- c. 150
- d. 180

MCQ 5. pandas .std() uses:

- a. $n-1$ (sample)
- b. n always
- c. $2n$
- d. zero

MCQ 6. SD is the square root of:

- a. Variance
- b. Mean
- c. Median
- d. n

Extra short questions

1. When mean vs median vs mode.
2. Population variance formula idea.
3. Why square root for SD?
4. Outlier trap with 9×10 and one 1000.

Extra long question

1. For 4,8,6,10,2 compute mean, median, range, population variance and SD. Then say what pandas std reports. Show every step. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Mean typical no outliers; median with outliers; mode popular category/value.

2. Mean of squared deviations; $\div n$. Sample $\div (n-1)$.
3. Back to original units.
4. Mean 109 vs median 10.

Long

1. Sorted 2,4,6,8,10. Mean 6, median 6, range 8. Devs -4,-2,0,2,4 sq 16,4,0,4,16 sum 40.
 $\sigma^2=8$, $\sigma=\sqrt{8}\approx 2.828$. pandas $40/4=10$, $\sqrt{10}\approx 3.162$.

Data and Analysis

8 single-page lessons · 5 Golden · read, close, say aloud, then sit the exam

Chapter in one look

- Data analysis turns raw facts into decisions. Python reads files and databases, calculates and plots.
- Connection chain: database → driver → connection → cursor → SQL.
- SQLite: connect, cursor, execute, commit, fetchall, close. No commit means no save.
- A DataFrame is a 2-D table. Load with a dictionary, `read_csv` or `read_excel`. New columns are vectorised arithmetic.
- NaN: find (`isnull`), fill (`fillna` , often with the mean), or drop (`dropna`).
- Tables give exact values; frequency bins show how many in each range.
- Bar/column = categories; line = time; pie = parts of a whole; histogram = numeric distribution (touching bars); scatter = two variables; box = five-number summary + outliers.
- Mean uses every value and is sensitive to outliers; median is the middle; mode is the most common.
- Range = max–min. Variance = mean squared deviation ($\div n$ or $\div n-1$). SD = $\sqrt{\text{variance}}$. pandas `.std()` uses $n-1$.

Key terms

Term	Meaning	اردو
Data analysis	Examining data to extract useful information.	ڈیٹا کا تجزیہ
Connectivity	Link between a program and a data source.	رابطہ کاری
Driver	Translator between Python and a DBMS.	ڈرائیور
Connection	Open pipeline to the database.	کنکشن
Cursor	Object that executes SQL and fetches rows.	کرسر
SQLite	Built-in file-based database.	ایس کیو لائٹ
commit	Save INSERT/UPDATE/DELETE to disk.	کمیت / محفوظ کرنا
Pandas	Library for table-like data.	پینڈا
DataFrame	2-D labelled table of rows and columns.	ڈیٹا فریم
NaN	Missing value ("Not a Number").	خالی / غیر موجود قدر
Frequency	How many items fall in a class/bin.	تعدد

Histogram	Distribution chart whose bars touch.	ہسٹوگرام
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CHAPTER 4 EXAM · CLOSED BOOK

Data and Analysis

75 marks · 1 hour 15 minutes · no notes · no phones

CHAPTER EXAM RULES

Instructions. Sit this exam after Lesson 08 and the recap. **Section A** — all 15 MCQs (1 mark). **Section B** — all 10 short questions (3 marks). **Section C** — attempt any **three** long questions (10 marks). At least one Section C answer must be a ★ Golden topic. Write in your notebook. Do not open the key until time is up.

Marks. 15 + 30 + 30 = 75. A 5-mark / 10-mark answer needs Define · Explain · Example · Diagram · Working.

سبق اور خلاصے کے بعد یہ امتحان بغیر نوٹس کے حل کریں۔ وقت پورا ہونے پر کلید کھولیں۔

Section A — Multiple choice (15 × 1 = 15)

- The object that executes SQL is the:
 - driver
 - cursor
 - DataFrame
 - histogram
- After INSERT you must:
 - close then commit
 - commit
 - drop the table
 - call head()
- Pandas is installed with:
 - import pandas
 - pip install pandas
 - sqlite3.connect
 - plt.show()
- `df.head()` shows:
 - the last 5 rows
 - the first 5 rows
 - only NaNs
 - the standard deviation
- Missing Pandas values appear as:
 - NULL in every chart
 - NaN
 - 0 always
 - a pie slice
- To delete every row that contains a gap:
 - fillna
 - dropna
 - fetchall
 - commit

7. A line graph is best for:
- a. parts of a whole
 - b. a trend over months
 - c. two unrelated categories
 - d. a primary key
8. Histogram bars:
- a. must have gaps
 - b. touch, because bins are a continuous range
 - c. always show time
 - d. replace the mean
9. Rainfall vs crop yield is a:
- a. pie chart
 - b. scatter plot
 - c. primary key
 - d. mode
10. A box plot's inner line is the:
- a. mean
 - b. mode
 - c. median
 - d. range
11. Mean of 70, 80, 90 is:
- a. 70
 - b. 80
 - c. 90
 - d. 240
12. Median of 10, 15, 35, 70, 95 is:
- a. 10
 - b. 35
 - c. 70
 - d. 45
13. Mode of 20, 25, 20, 30, 20, 25, 20 is:
- a. 25
 - b. 30
 - c. 20
 - d. 7
14. Range of 120, 150, 180 is:
- a. 30
 - b. 60
 - c. 150
 - d. 180
15. pandas `.std()` divides by:
- a. n
 - b. n-1
 - c. 2n
 - d. zero

Section B — Short questions (10 × 3 = 30)

1. List the five components of a Python database connection in order.
2. Why is `conn.commit()` required after INSERT?
3. What is a DataFrame? How do you load a CSV into one?
4. Differentiate `fillna` and `dropna`.

5. When is a pie chart a poor choice?
6. Differentiate a bar chart and a histogram.
7. Why is the median better than the mean for company salaries?
8. Define range, min and max with the list 70, 80, 90, 60, 75.
9. Why do we take the square root of variance?
10. State two careers that use this chapter's skills.

Section C — Long questions (attempt any $3 \times 10 = 30$)

1. Write a complete Python program that creates `school.db`, a `students` table, inserts Ali 85 and Sara 90, commits, prints all rows, and closes. Number the five connection components beside the matching lines.
2. Starting from the dictionary of Ali/Sara/Ahmed marks, write Pandas code to build a DataFrame, add Total and Average columns, and print it. Show the exact output.
3. Given Marks [85, None, 78, 92] and Grade ["A", "A", None, "A"], show `isnull().sum()`, then fill Marks with the mean and Grade with "Not Assigned". Compute the mean by hand.
4. For each goal pick a chart and justify in one sentence: (a) CS vs IT vs Eng enrolment, (b) website hits from Jan to Jun, (c) share of four clubs in one college, (d) distribution of 200 heights, (e) hours studied vs marks, (f) comparing spread of two classes.
5. For 4, 8, 6, 10, 2 compute mean, median, range, population variance and population SD. Show every step. Then say what pandas `.std()` would report instead.
6. Explain, with a numeric example, how an outlier distorts the mean but not the median. Link this to when you would draw a box plot.

Answer key. Finish the Chapter Exam first, then mark. Do not peek while the clock is running.

پہلے امتحان مکمل کریں، پھر اس کلید سے نمبر لگائیں۔

Check yourself — answers

- 4.1 Concept of Data Analysis.** Answers: 1. Study results to improve teaching (or similar). 2. Any two: read CSV/Excel/databases, organise tables, statistics, charts.
- 4.1.1 – 4.1.2 Data Sources and Database Connection.** Answers: 1. The cursor. 2. SQLite is the database; the `sqlite3` module is the driver.
- 4.2 Creating a SQLite Database in Python.** Answers: 1. All columns. 2. `conn.commit()`.
- 4.3 Basic Operations on Data using Pandas.** Answers: 1. The usual alias for pandas. 2. It would print everything; use `df.head()`.
- 4.3.3 Handling Missing Values (NaN).** Answers: 1. Count of NaNs in each column. 2. Any **row** that contains at least one NaN.
- 4.4 Data Organization and Representation.** Answers: 1. $4 + 2 = 6$. 2. Matplotlib (pyplot).
- 4.5 Data Visualization & Types of Graphs.** Answers: 1. The x-axis is a continuous numeric range, not separate categories. 2. Scatter (two numeric variables, looking for correlation).
- 4.6 Descriptive Statistics.** Answers: 1. Sorted 2,4,6,8,10 $\rightarrow$ 6. 2. 60. 3. $n-1$ (sample).

Answer key

Multiple choice

- | | | | | |
|------|------|------|-------|-------|
| 1. b | 4. b | 7. b | 10. c | 13. c |
| 2. b | 5. b | 8. b | 11. b | 14. b |
| 3. b | 6. b | 9. b | 12. b | 15. b |

Short questions — model answers

- Database, driver, connection object, cursor object, SQL queries.
- Without commit the insert stays in memory and disappears when the program ends, so the next run cannot see the row.
- A 2-D table of labelled rows and columns. `df = pd.read_csv("file.csv")`.
- `fillna(v)` replaces NaN with v (the row stays). `dropna()` deletes any row that still contains a NaN.
- When there are many slices, or the values are almost equal, so the eye cannot compare them.
- Bar: separate categories, gaps between bars. Histogram: one numeric variable grouped into bins; bars touch.
- A huge CEO salary pulls the mean up; the median stays with the typical worker.
- Min 60, max 90, range 30.
- Variance is in squared units. The SD brings the spread back to the original units (marks, rupees, °C).

10. Any two: data analyst, DBA, data scientist, visualisation specialist, BI analyst, backend developer.

Long questions — model answers

1. Use the nine-step listing in 4.2 with two INSERTs. Label: sqlite3 = driver, connect = connection, cursor = cursor, execute strings = SQL, school.db = database.
2. Output is the three-row table with Total 165, 178, 148 and Average 82.5, 89.0, 74.0 (see the worked example).
3. isnull sum: Marks 1, Grade 1. Mean of 85, 78, 92 = $255/3 = 85.0$. After fill: Sara Marks 85.0, Ahmed Grade "Not Assigned".
4. (a) column/bar — compare categories. (b) line — time. (c) pie — parts of a whole. (d) histogram — numeric distribution. (e) scatter — two numeric variables. (f) box — spread and outliers.
5. Sorted 2,4,6,8,10. Sum 30, mean 6, median 6, range 8. Deviations -4,-2,0,2,4. Squares 16,4,0,4,16 sum 40. $\sigma^2 = 40/5 = 8$. $\sigma = \sqrt{8} = 2.828$ (3 d.p.). pandas `.std()` uses $40/4 = 10$, $\sqrt{10} \approx 3.162$.
6. Nine salaries of 10 and one of 1000: mean 109, median 10. Draw a box plot — the 1000 appears as an outlier dot above the whisker, so the distortion is visible.

Self-assessment: I can...

- | | |
|--|--|
| ☐ name the five connection components in order. | ☐ write SQLite connect / create / insert / commit / select / close. |
| ☐ build a DataFrame and add a calculated column. | ☐ find, fill and drop missing values. |
| ☐ choose bar, line, pie, histogram, scatter or box for a problem. | ☐ compute mean, median, mode, range, variance and SD by hand. |
| ☐ state that pandas std uses $n-1$. | |